

Homework 5 Solutions

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Problem 12.1

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{4, 5, 6, 7\}$

- a) $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$.
- b) $A \cap B = \{4, 5\}$.
- c) $A - B = \{1, 2, 3\}$.
- d) $B - A = \{6, 7\}$.
- e) $A \Delta B = \{1, 2, 3, 6, 7\}$.
- f) $A \times B = \{(1, 4), (1, 5), (1, 6), (1, 7), (2, 4), (2, 5), (2, 6), (2, 7), (3, 4), (3, 5), (3, 6), (3, 7), (4, 4), (4, 5), (4, 6), (4, 7), (5, 4), (5, 5), (5, 6), (5, 7)\}$.
- g) $B \times A = \{(4, 1), (4, 2), (4, 3), (4, 4), (4, 5), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (7, 1), (7, 2), (7, 3), (7, 4), (7, 5)\}$.

Problem 12.2

Proof. If $|A| = 10$ and $|B| = 7$, then $|A \cup B| + |A \cap B| = 17$.

1. Observe that $|A \cup B| = |A| + |B| - |A \cap B|$.

2. Then $|A \cup B| + |A \cap B| = |A| + |B| = 10 + 7 = 17$. □

Problem 12.3

Proof. If $|A| = 10$ and $|B| = 7$, then $10 \leq |A \cup B| \leq 17$.

1. Observe that $|A \cup B| = |A| + |B| - |A \cap B| = 10 + 7 - |A \cap B| = 17 - |A \cap B|$.

2. Thus $|A \cup B|$ reaches a maximum when $|A \cap B|$ reaches a minimum, and vice versa.

3. Observe that $|A \cap B|$ reaches a minimum of 0 when sets A and B have no items in common, for example $A = \{1, 2, 3, \dots, 10\}$ and $B = \{11, 12, 13, \dots, 17\}$, in which case $|A \cup B| = 17$.

4. Observe that $|A \cap B|$ reaches a maximum when everything in B is also in A , for example $A = \{1, 2, 3, \dots, 10\}$ and $B = \{1, 2, 3, \dots, 7\}$, in which case $|A \cup B| = 10$.

5. Therefore $10 \leq |A \cup B| \leq 17$. □

Note: The x and y were stated in the proof with $x = 10$ and $y = 17$, and the corresponding examples being given.

Problem 12.5

Proof. $(A \cup B) \cup C = A \cup (B \cup C)$ and $(A \cap B) \cap C = A \cap (B \cap C)$.

1. Let A , B , and C be sets.

2. Then $(A \cup B) \cup C = \{x : x \in A \vee x \in B\} \cup C = \{x : (x \in A \vee x \in B) \vee x \in C\} = \{x : x \in A \vee (x \in B \vee x \in C)\} = A \cup \{x : x \in B \vee x \in C\} = A \cup (B \cup C)$.

3. Also $(A \cap B) \cap C = \{x : x \in A \wedge x \in B\} \cap C = \{x : (x \in A \wedge x \in B) \wedge x \in C\} = \{x : x \in A \wedge (x \in B \wedge x \in C)\} = A \cap \{x : x \in B \wedge x \in C\} = A \cap (B \cap C)$. □

Proof. $A \cup \emptyset = A$ and $A \cap \emptyset = \emptyset$.

1. Let A be a set.

2. Then $(A \cup \emptyset) = \{x : x \in A \vee x \in \emptyset\} = \{x : x \in A \vee \text{false}\} = \{x : x \in A\} = A$.

3. Also $(A \cap \emptyset) = \{x : x \in A \wedge x \in \emptyset\} = \{x : x \in A \wedge \text{false}\} = \{x : \text{false}\} = \emptyset$. □

Proof. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ and $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

1. Let A , B , and C be sets.

2. Then $A \cup (B \cap C) = A \cup \{x : x \in B \wedge x \in C\} = \{x : x \in A \vee (x \in B \wedge x \in C)\} = \{x : (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} = \{x : x \in A \vee x \in B\} \cap \{x : x \in A \vee x \in C\} = (A \cup B) \cap (A \cup C)$.

3. Also $A \cap (B \cup C) = A \cap \{x : x \in B \vee x \in C\} = \{x : x \in A \wedge (x \in B \vee x \in C)\} = \{x : (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C)\} = \{x : x \in A \wedge x \in B\} \cup \{x : x \in A \wedge x \in C\} = (A \cap B) \cup (A \cap C)$. □

Problem 12.7

Not worth the effort it would take to do this.

Problem 12.11

Proof. $A \cup B = A \cap B$ iff $A = B$.

1. ($\Rightarrow$) Let A and B be sets such that $A \cup B = A \cap B$.

2. Then everything in A or B is in both A and B .

3. Then $\forall x \in A, x \in A \cap B$ and therefore $x \in B$.

4. Then $\forall y \in B, y \in A \cap B$ and therefore $y \in A$.

5. Therefore $A = B$.

6. ($\Leftarrow$) Let A and B be sets such that $A = B$.

7. Then $A = A \cup A = A \cap A = A \cup B = A \cap B$. □

Problem 12.21

a) *Proof.* $A - (B - C) \neq (A - B) - C$.

1. Let $A = \{1\}$, $B = \{2\}$, and $C = \{1\}$.

2. Then $A - (B - C) = \{1\} - \{2\} = \{1\} \neq \emptyset = \{1\} - \{1\} = (A - B) - C$. □

b) *Proof.* $(A - B) - C = (A - C) - B$.

1. Let A , B , and C be sets, and let $a = x \in A$, $b = x \in B$, and $c = x \in C$.

2. Then $(A - B) - C = \{x : (a \wedge \neg b) \wedge \neg c\} = \{x : (a \wedge \neg c) \wedge \neg b\} = (A - C) - B$. □

c) *Proof.* $(A \cup B) - C \neq (A - C) \cap (B - C)$.

1. Let $A = \{1\}$, $B = \{2\}$, and $C = \{1\}$.

2. Then $(A \cup B) - C = \{1, 2\} - \{1\} = \{2\} \neq \emptyset = \emptyset \cap \{2\} = (A - C) \cap (B - C)$. □

d) *Proof.* If $A = B - C$, then it is not necessarily the case that $B = A \cup C$.

1. Let $A = \{1\}$, $B = \{1\}$, and $C = \{2\}$.

2. Then $A = \{1\} - \{2\} = B - C$, however $B = \{1\} \neq \{1, 2\} = A \cup C$. □

e) *Proof.* If $B = A \cup C$, then it is not necessarily the case that $A = B - C$.

1. Let $A = B = C = \{1\}$.

2. Then $B = \{1\} \cup \{1\} = A \cup B$, however $A \neq \emptyset = B - C$. □

f) *Proof.* $|A - B| \neq |A| - |B|$.

1. Let $A = \emptyset$, and let $B = \{1\}$.

2. Then $|A - B| = |\emptyset| = 0 \neq -1 = 0 - 1 = |A| - |B|$. □

g) *Proof.* $(A - B) \cup B \neq A$.

1. Let $A = \emptyset$, and let $B = \{1\}$.

2. Then $(A - B) \cup B = \emptyset \cup \{1\} = \{1\} \neq \emptyset = A$. □

h) *Proof.* $(A \cup B) - B \neq A$.

1. Let $A = B = \{1\}$.

2. Then $(A \cup B) - B = \{1\} - \{1\} = \emptyset \neq \{1\} = A$. □

Problem 12.22

a) *Proof.* $A = B$ iff $\overline{A} = \overline{B}$.

1. ($\Rightarrow$) Let A and B be sets such that $A = B$.

2. Let $x \in \overline{A}$.

3. Then $x \in \overline{A}$ implies $x \notin A$, and since $A = B$ then $x \notin B$ which gives $x \in \overline{B}$.

4. Therefore $\overline{A} \subseteq \overline{B}$.

5. Next let $x \in \overline{B}$.

6. Then $x \in \overline{B}$ implies $x \notin B$, and since $B = A$ then $x \notin A$ which gives $x \in \overline{A}$.

7. Therefore $\overline{B} \subseteq \overline{A}$.

8. Therefore $\overline{A} = \overline{B}$.

9. ($\Leftarrow$) The converse is identical to the first part, except switch A for $\overline{A}$ and switch B for $\overline{B}$. □

b) *Proof.* $\overline{\overline{A}} = A$.

1. Let A be a set, and let $x \in \overline{\overline{A}}$.

2. Since $x \in \overline{\overline{A}}$ then $x \in \overline{A}$. □

c) *Proof.* $(A \cup B) - C \neq (A - C) \cap (B - C)$.

1. Let $A = \{1\}$, $B = \{2\}$, and $C = \{1\}$.

2. Then $(A \cup B) - C = \{1, 2\} - \{1\} = \{2\} \neq \emptyset = \emptyset \cap \{2\} = (A - C) \cap (B - C)$. □

Problem 12.29

Proof. If A and B are finite sets then $|A \times B| = |A| \times |B|$.

1. Let A and B be finite sets.

2. Observe that $A \times B$ is the set of all ordered pairs where the first element is from A and the second element is from B .

3. Observe then that there are $|A|$ options for the first element, and that there are $|B|$ options for the second element.

4. Thus there are a total of $|A| \times |B|$ total ordered pairs in $A \times B$.

5. Therefore $|A \times B| = |A| \times |B|$. □

Problem 13.2

Proof. $(x - 1)(1 + x + x^2 + \dots + x^{n-1}) = x^n - 1$.

1. Observe that $(x - 1)(1 + x + x^2 + \dots + x^{n-1}) = x(1 + x + x^2 + \dots + x^{n-1}) - (1 + x + x^2 + \dots + x^{n-1}) = (x + x^2 + x^3 + \dots + x^n) - (1 + x + x^2 + \dots + x^{n-1})$.

2. Observe that every term cancels out except the x^n in the first expression and the 1 in the second expression.

3. Therefore $(x + x^2 + x^3 + \dots + x^n) - (1 + x + x^2 + \dots + x^{n-1}) = x^n - 1$.

4. Therefore $(x - 1)(1 + x + x^2 + \dots + x^{n-1}) = x^n - 1$. □

Proof. $2^0 + 2^1 + 2^2 + \dots + 2^{x-1} = 2^x - 1$.

1. Observe that $2^0 + 2^1 + 2^2 + \dots + 2^{x-1} = (1)(1 + 2^1 + 2^2 + \dots + 2^{x-1}) = (2 - 1)(1 + 2^1 + 2^2 + \dots + 2^{x-1})$.

2. Then using the previous proof and letting $x = 2$ we get $(2 - 1)(1 + 2^1 + 2^2 + \dots + 2^{x-1}) = 2^x - 1$.

3. Therefore $2^0 + 2^1 + 2^2 + \dots + 2^{x-1} = 2^x - 1$. □

Problem 13.4

Proof. Counting proof that if $a, b \in \mathbb{Z}^+$ where $a > b$, then $a^2 - b^2 = (a + b)(a - b)$.

1. Let $a, b \in \mathbb{Z}^+$ where $a > b$, and let $A = \{1, 2, 3, \dots, a\}$ and $B = \{1, 2, 3, \dots, b\}$.

2. Consider the set of ordered pairs $C = \{(x, y) : x \in A, y \in A, x > b \text{ or } y > b\}$, or in other words C is all the elements in $A \times A$ that contain an element greater than b .

3. Then C is all the elements in $A \times A$ that don't have both their elements less than or equal to b .

4. Observe that $B \times B$ is the set of all ordered pairs with both their elements less than or equal to b .
5. Then $C = A \times A - B \times B$, and since $B \times B \subseteq A \times A$, then $|C| = |A \times A| - |B \times B| = a^2 - b^2$.
6. Therefore $a^2 - b^2$ is the count of all the elements in C .
7. Another way of counting C is to count all the members whose first element is greater than b , and then all the elements whose first element is less than or equal to b but whose second element is greater than b .
8. Let $D = \{(x, y) \in C : x > b\}$, let $E = \{(x, y) \in C : x \leq b \text{ and } y > b\}$, and observe that $|C| = |D| + |E|$.
9. Observe that each element in D has $a - b$ options for the first element and a options for the second element.
10. Observe that each element in E has b options for the first element and $a - b$ options for the second element.
11. Then $|D| = a(a - b)$ and $|E| = b(a - b)$.
12. Then $|C| = |D| + |E| = a(a - b) + b(a - b) = (a + b)(a - b)$, showing that $a^2 - b^2$ and $(a + b)(a - b)$ both count the same set of objects C .
13. Therefore $a^2 - b^2 = (a + b)(a - b)$. □

Problem 13.5

Proof. $n^2 = n(n - 1) + n$ by counting.

1. Consider all the ordered pairs (x, y) that could be constructed from the set of numbers $\{1, 2, 3, \dots, n\}$.
2. Since we have n options for the x element and n options for the y element, then we have a total of n^2 possible outcomes.
3. Thus n^2 is the count of all possible ordered pairs, and it remains to show that $n(n - 1) + n$ is also the count.
4. Observe that in counting all the ordered pairs, we could first count all the pairs without any repeating elements and then count the pairs with repeating elements, thereby counting all the pairs.
5. Observe that when creating an ordered pair without any repetitions then we have n options for the first element, but only $n - 1$ options for the second (since we cannot choose the same number we used for the first element).
6. Thus $n(n - 1)$ is the count of ordered pairs without repeats.
7. Observe that when creating an ordered pair with only repetitions then we have n options for the first element, but only have 1 option for the second (since we have to choose the same number we used for the first element).
8. Thus $n \cdot 1 = n$ is the count of ordered pairs with repeats.
9. Thus $n(n - 1) + n$ is the count of all the ordered pairs.
10. Therefore, since both n^2 and $n(n - 1) + n$ count the same objects, then $n^2 = n(n - 1) + n$. □

Problem 13.7

Proof. $1 + 3 + 5 + \dots + (2n - 1) = n^2$ by counting.

1. Consider all the ordered pairs (x, y) that could be constructed from the set of numbers $\{1, 2, 3, \dots, n\}$.
2. Since we have n options for the x element and n options for the y element, then we have a total of n^2 possible outcomes.
3. Thus n^2 is the count of all possible ordered pairs.
4. It remains to show that $1 + 3 + 5 + \dots + (2n - 1)$ is also the count of all possible ordered pairs.
5. Observe that in counting all the ordered pairs we could first count all the pairs whose largest element is 1, then whose largest element is 2, ..., then whose largest element is n , thereby counting all the pairs.
6. Let a be an element between 1 and n , and consider the count of the set of ordered pairs A_a with a as the largest element.
7. In counting A_a , we can first count all the pairs where a is the first element and the second element is less than a (for a count of $1 \cdot (a - 1)$), then all the pairs where the first element is less than a and the second element is a (for a count of $(a - 1) \cdot 1$), and then finally the pair where both elements are a thereby counting all the pairs whose largest element is a .
8. Then the count of $|A_a| = 1 \cdot (a - 1) + (a - 1) \cdot 1 + 1 = 2(a - 1) + 1 = 2a - 2 + 1 = 2a - 1$.
9. Then the count of all pairs whose largest element is 1, then 2, then 3, ..., then n is $|A_1| + |A_2| + |A_3| + \dots + |A_n| = 1 + 3 + 5 + \dots + (2n - 1)$ is also the count of all ordered pairs.
10. Therefore $1 + 3 + 5 + \dots + (2n - 1) = n^2$. □